

Priya Aryav

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Summary

Software engineer with 6+ years building large-scale distributed systems, now working as a research-focused ML engineer on LLM pre-training, post-training, and inference. Builds and benchmarks training pipelines, an end-to-end chat model, and custom CUDA and Triton kernels from scratch.

Projects

- Building **teenychat**, an end-to-end ChatGPT-style assistant implemented from scratch in PyTorch, covering pre-training, post-training (SFT and RL), and inference, served through a web chat interface.
- Building **llmdotc**, an LLM training engine implemented from scratch in raw C/CUDA, and Triton, with custom kernels for core transformer operations, focused on maximizing GPU compute throughput for pre-training workloads.
- **tinyGPT**: Implemented four decoder-only transformer architectures from scratch in PyTorch (bigram, GPT, GPT-2, Qwen-2.5 with grouped query attention), with distributed training pipelines using DDP, mixed precision, gradient clipping, and cosine LR scheduling, benchmarking model FLOPs utilization across multi-GPU runs. Implemented key-value caching for efficient autoregressive inference and generation.
- Implemented **tinybpe**, a byte pair encoding tokenizer from scratch in Python, supporting basic and regex-based tokenization, special token registration, and model save and load functionality, validated by replicating OpenAI's GPT-4 tokenizer output exactly against production tiktoken results, with test coverage using pytest.
- Built **opengrad**, a minimal reverse-mode automatic differentiation engine supporting dynamic computation graphs and backpropagation, enabling training of small neural networks from first principles.
- Authored **technical write-ups** on transformer architecture, attention mechanisms, and tokenization, published on [Bear Blog](#).

Skills

Languages: **Python, C/C++, Java**
Frameworks: **PyTorch, CUDA, Triton**
ML Systems: **LLM Pre-training, Transformer Architectures, Inference optimization**
Systems: **Linux, Distributed Training (DDP), Performance Profiling**

Experience

Bengaluru, India

Lowe's | Software Engineer

Dec 2025 - Present

- Designed and implemented a distributed Java/Python-based data integration platform replacing Informatica and Mainframe, eliminating proprietary dependencies and saving \$5M annually through scalable architecture and performance optimization.

IBM | Software Engineer

Jun 2024 - Dec 2025

- Built real-time distributed data pipelines processing 2B+ records, reducing latency by 90%, and led adoption of Spock and Groovy testing frameworks across microservices, improving reliability by 30% and cutting production defects by 25%.

TCS | Software Engineer

Oct 2020 - Jun 2024

- Built Spring Boot microservices and Elasticsearch search from scratch, added features to existing services, and migrated 20+ legacy services (4x system efficiency, 50% higher Salesforce Bulk API throughput); cut response time by up to 50% and raised conversion by 14% through query optimization and caching, and reduced downtime by 80% with monitoring and alerting.

Education

B.Tech in Computer Science and Engineering

Aug 2016 - Jul 2020

Bengal College of Engineering and Technology, Durgapur

CGPA - 8.93